

FROM STRUCTURE TO SUBSTANCE
A NETWORK AND THEMATIC ANALYSIS OF THE HIP-HOP LYRICAL UNIVERSE

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Part I

THE RESEARCH QUESTION & METHODOLOGY

THE PROBLEM SPACE

- ▶ Hip-hop is often polarized by subjective labels: "Trap", "Conscious", "Melodic".
- ▶ **The Question:** Are these labels backed by data, or are they just marketing?
- ▶ **The Hypothesis (Homophily):** I posit that artists do not collaborate randomly.
- ▶ **Goal:** To mathematically quantify the link between **Social Structure** (Who you know) and **Lyrical Substance** (What you say).

DATA COLLECTION: A NOVEL APPROACH

Novel Data Source

I constructed a novel dataset using the **Genius API** to capture both collaboration metadata and full lyrical text.

Sampling Strategy: The "Four Poles"

- ▶ To avoid bias, I used 14 Seed Artists representing distinct sub-genres (Nas, Young Thug, Drake, Kanye West, etc.), **collecting the top 50 popular songs per artist.**
- ▶ **Method:** 1.5-Degree Breadth-First Search (BFS).

Final Scope: 180 Artists · 326 Connections · 700 Songs

METHODOLOGY: THE TWO-PRONGED PIPELINE

1. Structure (Network)

Graph Theory

- ▶ **Input:** Collaboration Graph
- ▶ **Metric:** Centrality (Degree & Betweenness)
- ▶ **Algorithm:** Louvain Community Detection

2. Substance (NLP)

Natural Language Processing

- ▶ **Input:** Raw Lyrics
- ▶ **Preprocessing:** Lemmatization + Custom Stopwords
- ▶ **Algorithm:** LDA Topic Modeling ($k = 5$)

Part II

FINDINGS: STRUCTURE, SUBSTANCE, AND SYNTHESIS

FINDINGS I: A SOLAR SYSTEM OF SOUND

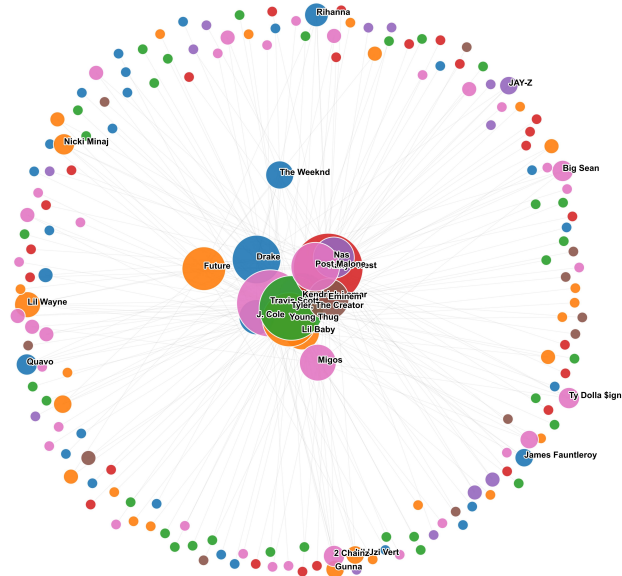


Figure. Network Visualization colored by Louvain Community.

THE SEVEN NATIONS OF HIP-HOP

The Louvain Algorithm identified 7 distinct communities. Here is the "Ground Truth" key for the rest of the analysis:

ID	Key Artist (Hub)	Genre Label	The "Vibe" (Description)
C0	Kendrick Lamar	Conscious / West Coast	<i>Social Commentary & Storytelling</i>
C1	Young Thug	Trap / Slime	<i>High Energy, Ad-libs, Money</i>
C2	Tyler, The Creator	Alternative	<i>Experimental & Genre-Bending</i>
C3	Kanye West	The Curators	<i>Producers who aggregate talent</i>
C4	Nas	The Purists	<i>Complex Lyricism & Tradition</i>
C5	Eminem	The Islanders	<i>Technical Rap, Pop Crossovers</i>
C6	Travis Scott	The Bridges	<i>Melodic Trap (Connects Pop & Rap)</i>

FINDINGS II: STRUCTURAL ROLES

The **Power Quadrant** separates "Curators" (High Volume) from "Bridges" (High Influence).

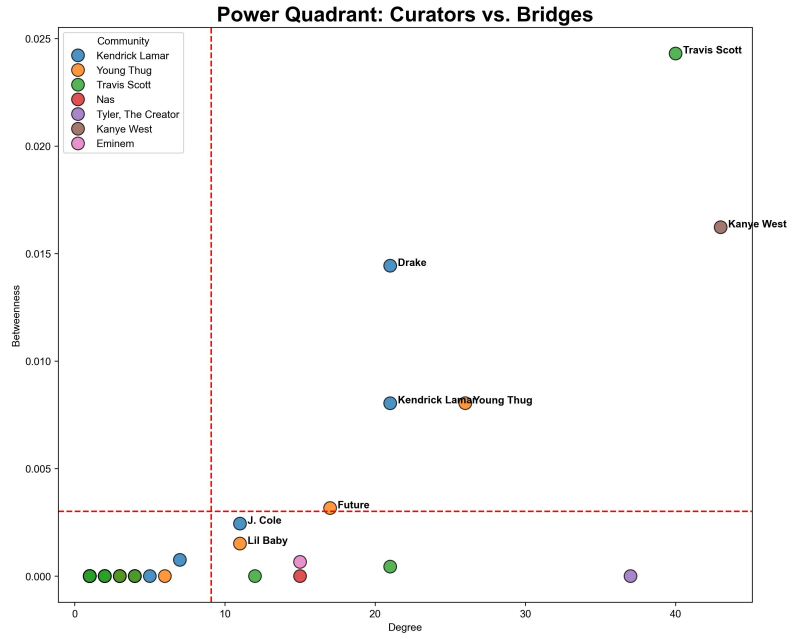


Figure. Scatter Plot: Degree vs. Betweenness Centrality

FINDINGS III: THE LYRICAL LANDSCAPE

The LDA model identified 5 distinct lyrical themes:



Topic 1 (Trap):
"Money", "Woo", "Gon"

Topic 3 (Conscious):
"Life", "God", "Real"

SYNTHESIS: STATISTICAL VALIDATION

Hypothesis Confirmation

A Chi-Squared test confirms a significant relationship between Social Community and Lyrical Topic.

$$(p < 6.36 \times 10^{-71})$$

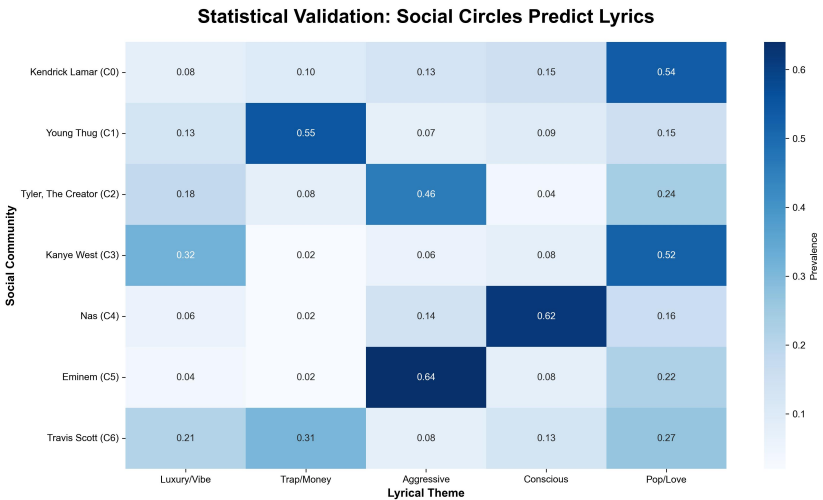


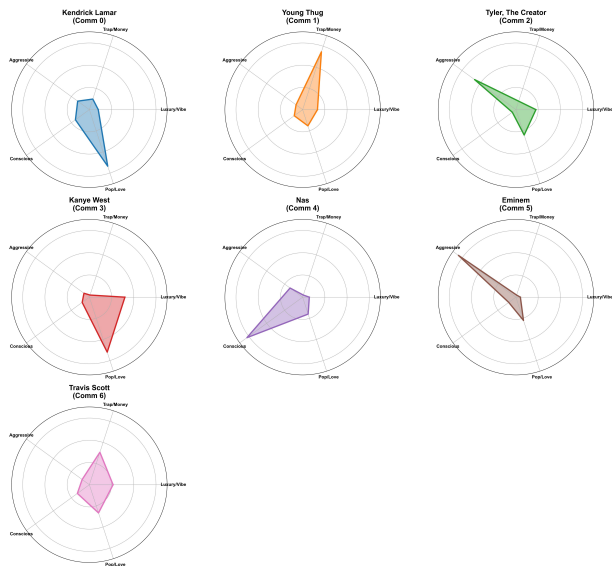
Figure. Heatmap: Community vs. Topic Prevalence

DEEP DIVE: COMMUNITY FINGERPRINTS

The Purist (Nas) *Highly Specialized (Spiky)*

The Bridge (Travis Scott) *Generalist (Round)*

Lyrical 'Fingerprints' by Community



Part III

CONCLUSION

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- ▶ The Hip-Hop industry is not a monolith; it is a system of **7 distinct "Nations."**
- ▶ **Homophily is Real:** Social isolation reinforces thematic echo chambers.
- ▶ **Bridges are Key:** Artists like Travis Scott maintain industry cohesion by being "multilingual."

Limitations & Future Work

- ▶ **Global Expansion:** Future work will incorporate multi-lingual datasets to analyze how these social structures function across different cultures and languages.
- ▶ **Sonic DNA:** I also aim to integrate **Audio Signal Processing** to determine if these clusters share distinct beat patterns and flows.

CONCLUSION

Thank You!

Questions?